Year 2 · Chapter 1 · Algebraic Methods

Starter

1. Factorise each polynomial.

(a) $x^2 - 6x + 5$

1a

(b) $x^2 - 16$

1b

(c) $9x^2 - 25$

1c

2. Simplify fully.

(a) $\dfrac{x^2 - 9}{x^2 + 9x + 18}$

2a

(b) $\dfrac{2x^2 + 5x - 12}{6x^2 - 7x - 3}$

2b

(c) $\dfrac{x^2 - x - 30}{-x^2 + 3x + 18}$

2c

3. Odd, even, or either?

For any integers n, m, decide whether each expression is always odd, always even, or could be either.

(a) $8n$

3a

(b) $n - m$

3b

(c) $3m$

3c

(d) $2n - 5$

3d

Apart from the initial portion on proof, this chapter concerns manipulation of algebraic fractions.

1: Proof By Contradiction

Prove by contradiction that $\sqrt{5}$ is an irrational number.

2: $+ \div \times -$ Algebraic Fractions

Express

$$\frac{3x+5}{x^2+x-12} - \frac{2}{x-3}$$

as a single fraction in its simplest form.

3: Express a fraction using partial fractions.

Express $\frac{2+3x}{(x+3)(x-1)}$ in the form $\frac{A}{x+3} + \frac{B}{x-1}$ where A and B are constants.

4: Divide algebraic expressions, and convert an improper fraction into partial fraction form.

Given that

$$\frac{x^3+x^2+7}{x-3} \equiv Ax^2 + Bx + C + \frac{D}{x-3}$$

find the values of A, B, C and D.

Prove that there is no greatest odd integer.

Assumption

Show contradiction

Conclusion

How to structure/word proof

"Assume that [negation of statement]."

[Reasoning followed by...] "This contradicts the assumption that..." or "This is a contradiction."

"Therefore [restate original statement]."

Proof by contradiction

Assume the statement is false. Prove this leads to a contradiction. The original statement must therefore be true.

Negating the original statement

Luisa · 5

The first part of a proof by contradiction requires you to negate the original statement. What is the negation of each statement? Click to choose.

There are infinitely many prime numbers.	**All Popes are Catholic.**	**If it is raining, my garden is wet.**
There are infinitely many non-prime (i.e. composite) numbers.	There exists a Pope who is not Catholic.	It is not raining and my garden is dry.
There are finitely many prime numbers.	No Popes are Catholic.	It is not raining and my garden is wet.
There are finitely many non-composite numbers.	Your Teacher is the Pope.	It is raining and my garden is not wet.

The negation of "all are" is "not all are", not "none are". For example, "not everyone likes green". Do not confuse negation with an opposite.

The negation of "if A then B" is "A and not B": the condition is true but the conclusion is false.

More Examples

Luisa · 6

Prove by contradiction that if n^2 is even, then n must be even.

Assumption

Show contradiction

Conclusion

Negating a conditional statement

The negation of "if A then B" is "A and not B".

Prove by contradiction that $\sqrt{2}$ is an irrational number.

Assumption

Show contradiction

Conclusion

Rational and irrational numbers

A rational number can be expressed as a/b, with integers a, b and $b \neq 0$. An irrational number cannot: examples include $\pi, e, \sqrt{3}$.

Rational numbers: $\mathbb{Q}$; real numbers: $\mathbb{R}$; natural numbers: $\mathbb{N}$; integers: $\mathbb{Z}$.

Alternative proof

In $2b^2 = a^2$, the power of 2 in the prime factorisation is odd on the left and even on the right. This is a contradiction.

Prove by contradiction that there are infinitely many prime numbers.

Assumption

Show contradiction

Conclusion

Euclid's proof

This is one of the earliest known proofs.

Each stage can be revealed separately: assumption, contradiction, then conclusion.

Exercise 5 Page 4 — Luisa · 9

5(a) Prove that if ab is irrational, then at least one of a, b is irrational. (3 marks)

5a

5(b) Prove that if $a + b$ is irrational, then at least one of a, b is irrational. (3 marks)

5b

Exercise 6 Page 4 — Luisa · 10

6. Use proof by contradiction to show that there exist no integers a, b for which $21a + 14b = 1$.

Working

Exercise 7a Page 4 — Luisa · 11

7(a) Prove by contradiction that if n^2 is a multiple of 3, n is a multiple of 3. (3 marks)

7(b) Hence prove by contradiction that $\sqrt{3}$ is irrational. (3 marks)

Working

Exercise 7b Page 4 — Luisa · 12

7(a) Prove by contradiction that if n^2 is a multiple of 3, n is a multiple of 3. (3 marks)

7(b) Hence prove by contradiction that $\sqrt{3}$ is irrational. (3 marks)

Working

Exercise 1A Page 4-5 — Luisa · 13

8. Use proof by contradiction to prove: "There are no integer solutions to $x^2 - y^2 = 2$."

Working

Exercise 9 Page 4 — Luisa · 14

9. Prove by contradiction that $\sqrt[3]{2}$ is irrational.

Working

Extension

[STEP I 2008 Q1] What does it mean to say that a number x is rational?
Prove by contradiction statements A and B below, where p and q are real numbers.
A: If pq is irrational, then at least one of p and q is irrational.
B: If $p+q$ is irrational, then at least one of p and q is irrational.
Disprove by means of a counterexample statement C below, where p and q are real numbers.
C: If p and q are irrational, then $p+q$ is irrational.
If the numbers e, π, π^2, e^2 and $e\pi$ are irrational, prove that at most one of the numbers $\pi+e, \pi-e, \pi^2-e^2, \pi^2+e^2$ is rational.
Solutions on next slide.

Note: e is Euler's Number and you will encounter it later at A Level. But for the moment, you only need to know it is an irrational number, like π.

Solution to Extension Problem **Luisa · 16**

A number x is rational if $x = m/n$ for integers m, n with $n \neq 0$. An irrational number cannot be written this way.

A. If pq is irrational, at least one of p, q is irrational.

A

B. If $p+q$ is irrational, at least one of p, q is irrational.

B

C. Disprove: if p, q are irrational, then $p+q$ is irrational.

C

Solution to Extension Problem

Assuming $e, \pi, \pi^2, e^2, e\pi$ are irrational, prove that at most one of $\pi + e, \pi - e, \pi^2 - e^2, \pi^2 + e^2$ is rational.

2a: Multiplying/Dividing Algebraic Fractions

Multiply algebraic fractions as you would numerical fractions. Factorise and cancel common factors where possible.

$$\frac{a}{b} \times \frac{c}{a} =$$

$$\frac{x+1}{2} \times \frac{3}{x^2-1} =$$

To divide by a fraction, multiply by the reciprocal of the second fraction.

$$\frac{p}{q} \div \frac{r}{q} =$$

$$\frac{x+2}{x+4} \div \frac{3x+6}{x^2-16} =$$

Test Your Understanding — Luisa · 19

$$\frac{x+3}{5} \times \frac{10}{x^2-9} =$$

$$\frac{x^2+x}{y} \div \frac{x^2-x-2}{y^2} =$$

Spot the error

A student writes $\frac{x^2+y}{2y} = \frac{x^2}{2}$. Explain why this is incorrect.

Explanation

Exercise 1B (page 6-7) — Luisa · 20

3. Show that

$$\frac{x^2-64}{x^2-36} \div \frac{64-x^2}{x^2-36} = -1.$$

Working

4. Show that

$$\frac{2x^2-11x-40}{x^2-4x-32}\times\frac{x^2+8x+16}{6x^2-3x-45}\div\frac{8x^2+20x-48}{10x^2-45x+45}=\frac{a}{b},$$

and find the integer constants a, b.

Working

2b: Adding/Subtracting Algebraic Fractions **Luisa · 22**

To add or subtract fractions, find a common denominator.

Simplify $\frac{3}{x+1}-\frac{2}{x+2}$ and $\frac{3}{x+1}-\frac{4x}{x^2-1}$.

Working

Edexcel C3 June 2013(R), Q1

Express as a single fraction in its simplest form:

$$\frac{3x+5}{x^2+x-12}-\frac{2}{x-3}.$$

1

Express as a single fraction in its simplest form:

$$\frac{10x+4}{3x^2+4x+1}-\frac{3}{x+1}.$$

Exercise 1C · Pure Mathematics Year 2

Textbook page 8, Chapter 1: Algebraic Methods.

Open Exercise 1C in the textbook

Complete the exercise. Use the techniques for adding and subtracting algebraic fractions from the preceding examples.

Partial Fractions

If the **denominator is a product of a linear terms**, it can be split into the sum of 'partial fractions', where **each denominator is a single linear term**.

$$\frac{6x-2}{(x-3)(x+1)} \equiv \frac{A}{x-3} + \frac{B}{x+1}$$

Notation reminder: $\equiv$ means 'equivalent/identical to', and indicates that both sides are equal for **all** values of x.

Method 1: Substitution

Method 1

Method 2: Comparing Coefficients

Method 2

Further Example

Given that

$$\frac{6x^2+5x-2}{x(x-1)(2x+1)} \equiv \frac{A}{x} + \frac{B}{x-1} + \frac{C}{2x+1},$$

find A, B, C.

Working

Test Your Understanding — Luisa · 27

Edexcel C4 June 2005, Q3(a)

Express $\dfrac{5x+3}{(2x-3)(x+2)}$ in partial fractions. (3 marks)

Working

Exercise 1D page 8 — Luisa · 28

4. Express $\dfrac{6x+1}{x^2+2x-15} - \dfrac{4}{x-3}$ as a single fraction in its simplest form.

Working

6. Express $\dfrac{4(2x-1)}{36x^2-1}+\dfrac{7}{6x-1}$ as a single fraction in its simplest form.

Working

7. $g(x)=x+\dfrac{6}{x+2}+\dfrac{36}{x^2-2x-8}$, for $x\in\mathbb{R}$, $x\neq -2, 4$.

(a) Show that $g(x)=\dfrac{x^3-2x^2-2x+12}{(x+2)(x-4)}$.

7a

(b) Using algebraic long division, or otherwise, show that $g(x)=\dfrac{x^2-4x+6}{x-4}$.

7b

Repeated linear factors

Luisa · 31

Suppose we wished to express $\frac{2x+1}{(x+1)^2}$ as $\frac{A}{x+1}+\frac{B}{x+1}$. What's the problem?

Explanation

Split $\frac{11x^2+14x+5}{(x+1)^2(2x+1)}$ into partial fractions.

$$\frac{11x^2+14x+5}{(x+1)^2(2x+1)} \equiv \frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{2x+1}$$

The problem is resolved by having the factor **both squared and non-squared** (explanation of why we do this at the end of these slides).

Working

Test Your Understanding

Luisa · 32

Edexcel C4 June 2011, Q1

Given $\dfrac{9x^2}{(x-1)^2(2x+1)} \equiv \dfrac{A}{x-1}+\dfrac{B}{(x-1)^2}+\dfrac{C}{2x+1}$, find A, B, C. (4 marks)

Working

Exercise 1E Year 2 page 13 — Luisa · 33

1. $f(x) = \dfrac{3x^2 + x + 1}{x^2(x+1)}$, $x \neq 0, -1$.

Given $f(x) \equiv \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x+1}$, find A, B, C.

Working

Exercise 1E Year 2 page 13 — Luisa · 34

4. Show that $\dfrac{5x^2 - 2x - 1}{x^3 - x^2}$ can be written as $\dfrac{C}{x} + \dfrac{D}{x^2} + \dfrac{E}{x-1}$, finding C, D, E.

Working

Dealing with Improper Fractions

Luisa · 35

In Pure Year 1, we saw that the '**degree**' of a polynomial is the highest power, e.g. a quadratic has degree 2.
An algebraic fraction is **improper** if the degree of the numerator is **at least** the degree of the denominator.

$\frac{x^2-3}{x+2}$ $\frac{x+1}{x-1}$ $\frac{x^3-x^2+3}{x^2-x}$

A partial fraction is still improper if the degree is the same top and bottom.

Questions might take one of two forms:
- Do the division to express as a quotient and a remainder, e.g. $\frac{x+1}{x-1} \quad \rightarrow \quad 1+\frac{2}{x-1}$
- Express as partial fractions, e.g. $\frac{x^2+x}{(x+1)(x-2)} = A + \frac{B}{x+1} + \frac{C}{x-2}$

Reducing to Quotient and Remainder

Luisa · 36

For example, $7 \div 3 = 2$ remainder 1, so $\frac{7}{3} = 2 + \frac{1}{3}$.

In general,

$$\frac{F(x)}{\text{divisor}} = Q(x) + \frac{\text{remainder}}{\text{divisor}},$$

where $Q(x)$ is the quotient.

If

$$\frac{x^2+5x-9}{x+2} = Ax + B + \frac{C}{x+2},$$

determine A, B, C.

Division

Result

Test Your Understanding — Luisa · 37

Given that

$$\frac{3x^4 - 2x^3 - 5x^2 - 4}{x^2 - 4} \equiv ax^2 + bx + c + \frac{dx + e}{x^2 - 4}, \quad x \neq \pm 2,$$

find the constants a, b, c, d, e. (4 marks)

Tip: insert $+0x$ for missing powers before dividing.

Division

Alternative

Dealing with Improper Fractions — Luisa · 38

Split $\frac{3x^2-3x-2}{(x-1)(x-2)}$ into partial fractions.

Method 1: Algebraic Division

Method 2: Using One Identity

Method 1

Method 2

Edexcel C4 January 2013, Q3

Express $\dfrac{9x^2 + 20x - 10}{(x+2)(3x-1)}$ in partial fractions. (4 marks)

Working

6. Given $\dfrac{2x^2 - 1}{x^2 + 2x + 1} \equiv A + \dfrac{B}{x+1} + \dfrac{C}{(x+1)^2}$, find A, B, C. (4 marks)

Working

7. By factorising the denominator, express each fraction in partial fractions.

(a) $\dfrac{4x^2+17x-11}{x^2+3x-4}$

(b) $\dfrac{x^4-4x^3+9x^2-17x+12}{x^3-4x^2+4x}$

7a

7b

8. Given $\dfrac{6x^3-7x^2+3}{3x^2+x-10} \equiv Ax+B+\dfrac{C}{3x-5}+\dfrac{D}{x+2}$, find A, B, C, D. (6 marks)

Working

If a factor is repeated, why do have a partial fraction with the squared and one without?
When we split into partial fractions, we want each fraction to be non-top-heavy algebraic fractions – recall this means that the 'order' of the numerator has to be less than the order of the denominator. We assume the most generic non-top-heavy fraction possible, i.e. where the order of the numerator is one less than the denominator…

order 1

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2}$$

order 2

Since the denominator is quadratic (order 2), we want the **most generic possible numerator whilst avoiding the fraction being top heavy**, i.e. linear (order 1). Just putting $\frac{B}{x^2}$ wouldn't be good enough because there *might* have been an x term in the numerator.

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{Bx}{x^2} + \frac{C}{x^2}$$

Split the fraction.

We can see therefore we have both the x with and without the squared in the denominator.

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{B}{x} + \frac{C}{x^2}$$

This is easy enough if the repeated factor is x^2, but what about more general repeated linear factors?

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{Bx+C}{(x+1)^2}$$

We want something in the numerator where a $(x+1)$ will cancel, so a bit of clever manipulation is required.

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B(x+1)-B+C}{(x+1)^2}$$

Split the fraction as before.

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B(x+1)}{(x+1)^2} + \frac{C-B}{(x+1)^2}$$

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{D}{(x+1)^2}$$

$C-B$ is just a generic constant, so replace with a single constant.